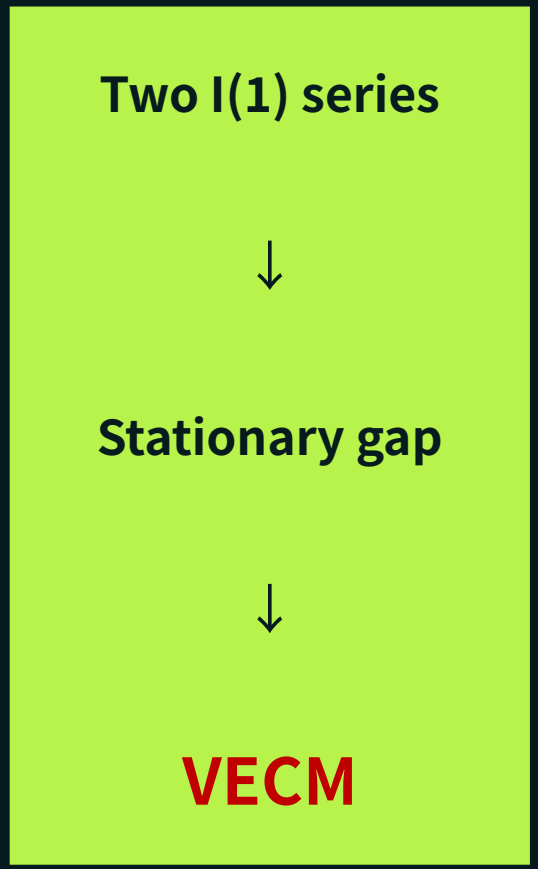
Cointegration & the VECM
The Pull to Equilibrium

How two wandering series are tied to the same equilibrium



Financial Time Series · 07

We build up cointegration and the VECM in five steps

01	Basics	**Nonstationarity and spurious regression** I(1) · I(0) · pros and cons of differencing
02	Cointegration and EG test	**A stationary gap means cointegration** Definition · intuition · Engle–Granger two-step
03	VECM and Johansen	**Modeling the adjustment that closes the gap** Adjustment speed · half-life · rank
04	Causality and pairs trading	**Long-run causality in practice** Granger theorem · two causalities · ±2σ
05	Cases and lab	**Equilibrium, but no timing promise** Rates · dividends · basis · LTCM · Samsung

Basics → definition & test → model → causality & strategy → cases & lab

A random walk is nonstationary, but one difference makes it stationary

Basics — the order of integration I(d) is the number of differences needed for stationarity

Random walk — the level is I(1)

$$y_t = y_{t-1} + e_t \quad \Rightarrow \quad y_t \sim I(1)$$

Difference — the change is I(0)

$$\Delta y_t = y_t - y_{t-1} = e_t \sim I(0)$$

I(1) · nonstationary

Shocks accumulate

never reverts to a mean

I(0) · stationary

Constant mean, variance

reverts to the mean

Difference Δ

This period – last

turns I(1) into I(0)

Cointegration asks whether a combination of I(1) series is I(0)

Regressions in levels and in differences each have a flaw

Basics — the dilemma when regressing one I(1) variable on another

Regression in levels

Spurious regression

- Unrelated I(1) series
- look significant

the link may be fake

Cointegration

solves both problems

Differences only

Long-run info lost

- Only changes remain
- levels discarded

equilibrium is invisible

Cointegration identifies when a level relation is genuine

Short-run dynamics and the long-run equilibrium fit in one model

Two I(1) series are cointegrated if a linear combination is I(0)

Definition by Engle & Granger (1987)

Definition of cointegration

$$y_t,\ x_t \sim I(1), \qquad y_t - k\,x_t \sim I(0) \qquad (1,\ -k):\ \text{cointegrating vector}$$

Long-run slope k

Equilibrium slope

pairs-trade hedge ratio

Gap · ECT

Disequilibrium

reverts to its mean

Cointegrating vector

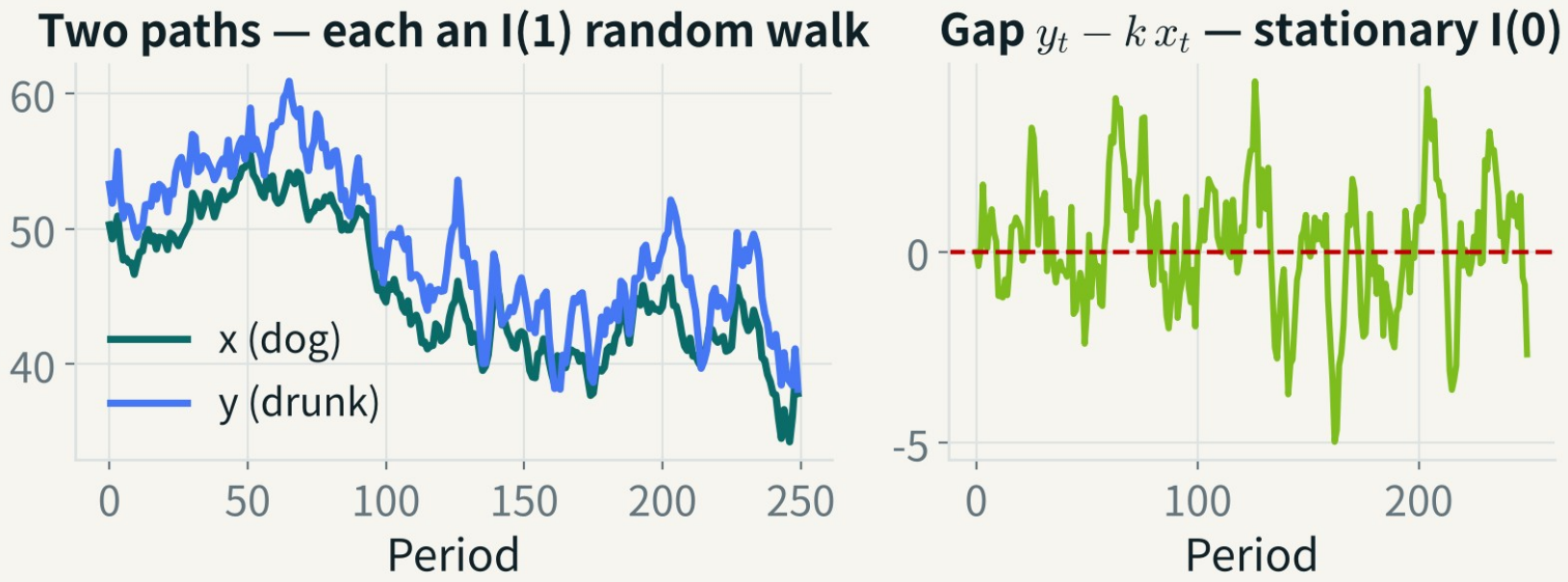
(1, –k)

weights forming the gap

Each series wanders, yet the gap stays within a range

For two leashed random walks, only the gap is stationary

The drunk and her dog (Murray, 1994) — both wander, but stay within the leash

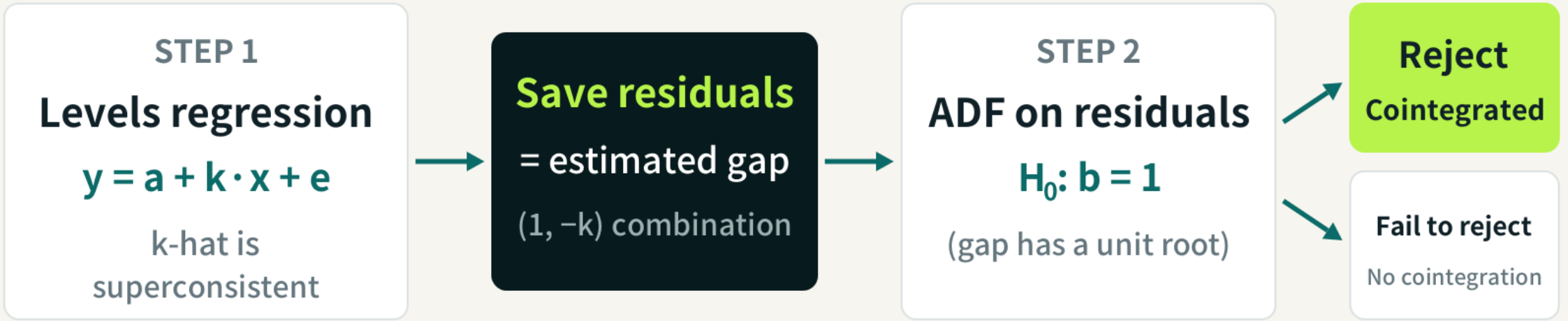


※ Simulated data (seed 42) — x is a random walk; the gap is AR(1) with coefficient 0.8; k = 1

The paths are unpredictable, but the gap's reversion is predictable

The Engle–Granger test is one regression plus one unit-root test

A two-step procedure suited to the bivariate case



Critical value: Engle–Granger (MacKinnon) ≈ −3.34, not the usual ADF −2.86

If the residuals have no unit root, the gap is stationary and the pair is cointegrated

Step 2 checks whether the residual persistence b is below 1

b is the share of the gap left next period; the d terms absorb autocorrelation

STEP 1 — levels regression and residuals

$$y_t = a + k\,x_t + e_t \quad \Rightarrow \quad \hat{e}_t = y_t - \hat{a} - \hat{k}\,x_t$$

STEP 2 — ADF regression on the residuals

$$\Delta\hat{e}_t = (b-1)\,\hat{e}_{t-1} + d_1\Delta\hat{e}_{t-1} + \cdots + d_q\Delta\hat{e}_{t-q} + u_t, \qquad H_0:\ b = 1$$

※ Residuals are estimated: use Engle–Granger (MacKinnon) critical values, ≈ −3.34 at 5% (ADF: −2.86)

Limits: results depend on which variable is on the left; at most one relation is assumed

Rejecting b = 1 means the gap shrinks — cointegration

A VECM is a VAR in differences plus the lagged gap

Vector error-correction model — the bivariate case

y equation

$$\Delta y_t = a_y \underbrace{(y_{t-1} - k\,x_{t-1})}_{\text{ECT}} + b_1 \Delta y_{t-1} + c_1 \Delta x_{t-1} + e_t$$

x equation

$$\Delta x_t = a_x \underbrace{(y_{t-1} - k\,x_{t-1})}_{\text{ECT}} + \cdots + e_t^*$$

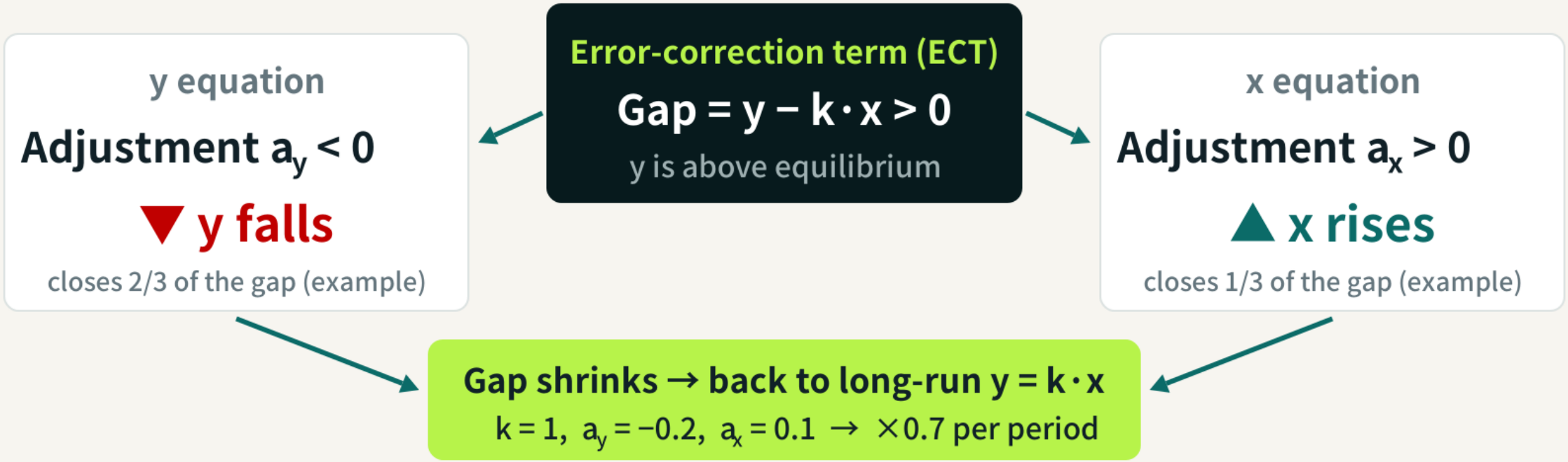
How to read

ECT	lagged gap
k	long-run slope
a	adjustment speed
b · c	short-run terms

Short-run dynamics and the long-run equilibrium sit in one equation

When the gap is positive, y falls and x rises

The sign of the adjustment speed sets the direction — negative in the y equation, positive (or 0) in x



Adjustment speeds give the size and direction of the pull to equilibrium

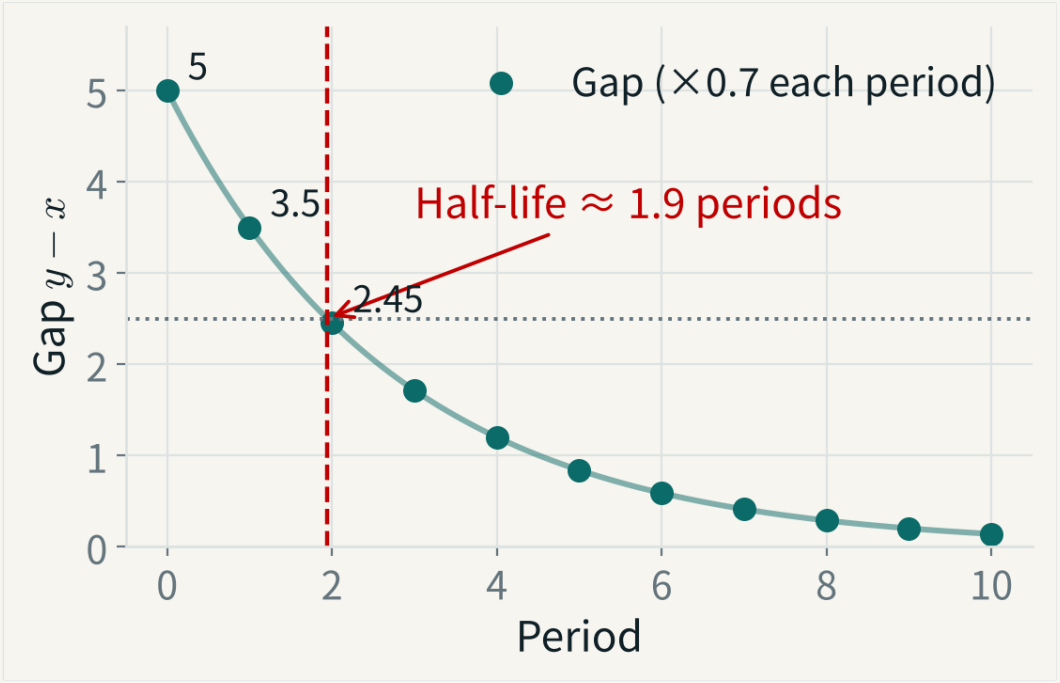
The gap shrinks by 0.7× each period and halves in about 1.9 periods

Example — k = 1, adjustment −0.2 (y) and 0.1 (x), initial gap 5, no short-run terms or shocks

Reading the result

- 5 → 3.5 → 2.45
- ×0.7 each step
- y 2/3 · x 1/3

half-life ≈ 1.9



The variable with the larger |adjustment speed| moves more toward equilibrium

The number of cointegrating relations determines the right model

With N variables, the choice depends on the cointegrating rank r

r = 0	0 < r < N	r = N
No cointegration no long-run relation	**r long-run equilibria** short run + long run	**All effectively I(0)** no differencing needed
↓	↓	↓
VAR in differences	**VECM**	**VAR in levels**

Using a VAR in differences under cointegration throws away the long-run relation

In the general form, the rank of C is the number of relations

Reference — matrix form with N variables stacked in a vector z (advanced)

VECM in matrix form

$$\Delta z_t = C\, z_{t-1} + D_1 \Delta z_{t-1} + \cdots + D_{p-1} \Delta z_{t-p+1} + \text{const} + e_t$$

C written out for two variables

$$C = \underbrace{\begin{pmatrix} a_y \\ a_x \end{pmatrix}}_{\text{loading}} \underbrace{\begin{pmatrix} 1 & -k \end{pmatrix}}_{\text{cointegrating vector}'}, \qquad \text{rank}(C) = r$$

Notation

C	long-run matrix
D	short-run coefficients
r	rank C = # relations

C combines adjustment speeds and the cointegrating vector — both equations follow

The Johansen test asks about the rank sequentially, starting from zero

Estimate the VECM by maximum likelihood and infer r from the eigenvalues of C, one candidate at a time

$$\text{Trace:} \quad H_0 : \; r \leq r_0 \quad \text{vs} \quad H_1 : \; r > r_0$$

$$\text{Max-eigenvalue:} \quad H_0 : \; r = r_0 \quad \text{vs} \quad H_1 : \; r = r_0 + 1$$

① r = 0?
Rejected

→

② r ≤ 1?
Not rejected

→

Conclude
r = 1

Critical values depend on whether constants and trends enter inside or outside the relation

The first r that is not rejected is the cointegrating rank

In a VECM, long-run and short-run causality are tested separately

Granger representation theorem — cointegration implies causality in at least one direction

$$\Delta y = \underbrace{a_y \times (\text{last gap})}_{\text{error-correction term}} + \underbrace{c_1 \times \text{lagged } \Delta x}_{\text{short-run dynamics}} + \cdots + e$$

Long-run causality

Test: $a_y = 0$

if zero, y is weakly exogenous

Short-run causality

Test: $c_1 = 0$

effect of lagged changes

A VAR in differences drops the ECT and misses long-run causality

※ Alternative: Toda–Yamamoto — VAR in levels with extra lags; Wald-test only the original lags

Under cointegration, at least one adjustment speed is nonzero

Pairs trading bets on mean reversion of the spread

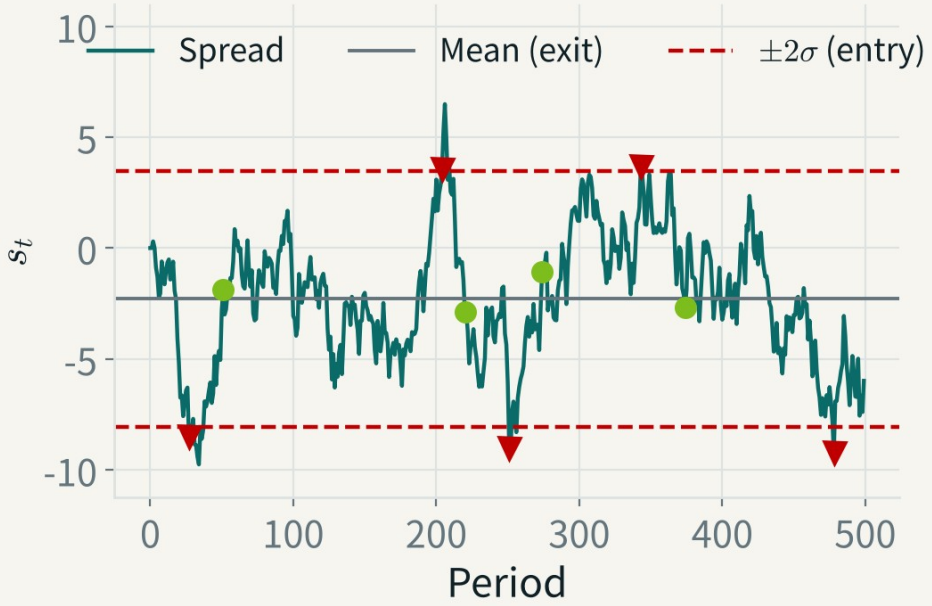
Two stocks in one industry — enter when the spread built with hedge ratio k moves beyond ±2σ

$$s_t = P_{A,t} - \hat{k}\, P_{B,t}, \qquad \text{entry: } |s_t - \bar{s}| \geq 2\sigma$$

Half-life

$$\text{half-life} = \frac{\ln 0.5}{\ln b}$$

b = 0.9:
gap halves in
≈ 6.6 periods



※ Simulated data (seed 7) — ▼ entry · ● exit

M&A and business changes break pairs — use rolling tests, stop-losses, out-of-sample checks

Testing many pairs mechanically produces spurious cointegration

Rates, dividends and the basis have theory-given equilibria

Cointegration with economic grounds — expectations theory, present-value model, cost-of-carry arbitrage

Expectations theory

Long vs short rates

$$i_t^L - i_t^S \sim I(0)$$

A wide spread signals short-rate moves

Episode

2022–23 US curve inversion — recession-signal debate

Present-value model

Price and dividend

$$\ln P_t - \ln D_t \sim I(0)$$

Future returns or dividends close yield gaps

Episode

2000 dot-com peak: record-low yield → weak decade

Cost of carry · arbitrage

Spot and futures

$$F_t - S_t\,(1+c) \sim I(0)$$

Basis beyond costs invites arbitrage

Episode

Nov 2010 expiry — an arbitrage unwind sank KOSPI

Apply cointegration tests only to pairs with an economic rationale

An equilibrium does not guarantee when convergence happens

Twin shares Royal Dutch / Shell — fixed 60:40 split, theoretical price ratio 1.5 (Froot & Dabora, 1999)

Log price ratio that should be stationary in theory

$$\ln V_t^{RD} - \ln V_t^{Shell} - \ln 1.5 \sim I(0)$$

1907

60:40 alliance

fair ratio 1.5

1980s–90s

Gap persists

never hits 1.5

1998 · LTCM

Gap widens

big losses

2005

Unified company

gap disappears

Limits to arbitrage — without staying power, you are forced out before convergence

Cointegration tells you the direction, not the timing

The common–preferred gap mean-reverts, but the mean itself moves

Samsung Electronics 005930 · 005935 — two claims on the same cash flows

Gap (log price difference)

$$\ln P_t^{com} - \ln P_t^{pref} \sim I(0)$$

Common ground

Same cash flows

move together long run

Differences

Rights & liquidity

set the mean gap

Episode · 2018

50-for-1 split

ratio unchanged

Read the gap's half-life together with dividend and liquidity differences

Claude Code tests cointegration and fits a VECM to rates

Lab 1 — FRED DGS10 · DGS3MO, monthly averages from 1990

① ADF · KPSS confirm I(1) → ② EG · Johansen rank → ③ Fit VECM k · adjustment → ④ Spread ±2σ · half-life → ⑤ md report open w/ figures

Delegate the reading too

Who adjusts more?

long vs short rate

Your role

Paste and check

Claude Code downloads, runs, fixes

Output: vecm_yield_curve.md — ask Claude Code to open it with the figures

The common–preferred pairs trade is validated out of sample

Lab 2 — yfinance 005930.KS · 005935.KS, daily adjusted closes from 2012

① Log prices EG test → ② 252-day roll stability → ③ Enter z ±2 exit near 0 → ④ 0.1% cost out-of-sample → ⑤ md report open w/ figures

Performance

Return · max drawdown

plus trade count, in a table

Validation design

Fit 5 yrs, test after

does it hold out of sample?

Let Claude Code fix errors itself and carry on to the end

EACH SERIES WANDERS; THE GAP RETURNS.

ADJUSTMENT SPEEDS SHOW WHO MOVES.

CONVERGENCE TIMING IS NOT GUARANTEED.

Read in order: cointegration test →
adjustment speed → half-life.